COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S2
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ASSIGNMENT NO. 6.1 Linear and Binary Search

1. PROCEDURE OUTPUT

What procedures did you perform?

1. What is a search tree in data structures?

A search tree is a node-based and hierarchical data structure designed to efficiently search, insert, and delete data. It arranges data in an organized order where operations such as searching can be performed without slowly searching through the entire tree to form a more efficient data retrieval operation. A search tree consists of a main tree, the values, a left subtree, and the right subtree, which are the main parts of this data structure. It follows an order where the values smaller than the main tree are placed at the left subtree and those with smaller values are placed at the right. By using this tree layout, a search operation can bypass half of the tree and take a smaller amount of time to search a node.

2. What are the Different types of search algorithm in data structures? Differentiate each type of search.

The following different types of search algorithms in data structures are Sequential and Interval search. Through linear search, it searches through the elements individually in a sequential method from the start to the end of the list. This search method works on both sorted and unsorted structures. But since it searches through the data one by one, it gets slower as the size of the list becomes larger.

In contrast to sequential search, interval search requires sorted data since it divides the range of data into segments, which cuts down most of the data in the search operation. Binary search is a fast method that uses the sorted data to look for the middle values and compare them to the target value, and then discard the rest of that node until the target value is reached. Ternary search is similar to binary search in terms of its methods but rather than using two subtrees, it uses three subtrees to compare the values and discards the other two if the target value or range is reached. Exponential search looks through a list by finding the range where the element is present with the square of the index, and performs a standard binary search in the selected range.

3. What operations / implementations can be performed using binary and linear search operations?

Linear search is ideal for unorganized lists because this operation searches through each item individually. Because of the simplicity of this operation, it can easily find the smallest and largest values, count occurrences of the elements, and mark specific values in the lists such as a missing, duplicate, and invalid type. Meanwhile, binary search operations require lists that are sorted since it halves the list at each step, which means it can find the first and last occurrence of a value. It can also find values at a certain range, find the highest or lowest value, and insert or delete a value inside a specified index.

4. What are the advantages in using binary search tree as data structure?

The following advantages in using a binary search tree are: Resizable structure, more efficient data operations, and automatic sorting. A resizable structure means that a binary search tree can dynamically expand depending on the amount of nodes unlike arrays which have a fixed limit defined in the code and require removing elements if the limit is reached. It has more efficient data operations since the data subtrees are separated into two halves where the smaller values go to the left and the larger ones to the right, so the search operation takes half of that for a full search. Since this type of data structure is a hierarchical tree where data is automatically organized by values smaller and larger than the main tree, sorting is a simple process and does not require elaborate algorithms to get the desired order.

5. Give an example program using binary search and Linear search.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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